Course Code: PH7302E **Session:** Winter Semester 2026–27
Course Title: Condensed Matter Physics **Maximum Marks:** 50
Evaluation Split: Pre-Midterm 30% (15 Marks) Post-Midterm 70% (35 Marks)

Physical Constants: $h = 6.626 \times 10^{-34}$ J·s, $\hbar = 1.055 \times 10^{-34}$ J·s, $e = 1.602 \times 10^{-19}$ C, $m_e = 9.109 \times 10^{-31}$ kg, $k_B = 1.381 \times 10^{-23}$ J·K^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1}, $\epsilon_0 = 8.854 \times 10^{-12}$ F·m^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H·m^{-1}, $\mu_B = 9.274 \times 10^{-24}$ J·T^{-1}, $c = 2.998 \times 10^8$ m·s^{-1}, 1 eV $= 1.602 \times 10^{-19}$ J, 1 amu $= 1.661 \times 10^{-27}$ kg.

PART A: PRE-MIDTERM SOLUTIONS [15 MARKS]

Question 1: Crystal Structure & X-ray Diffraction [7 Marks]

(a) Structure Factor for Face-Centered Cubic (FCC) Lattice [4 Marks]

(i) **Definition:** The geometric structure factor F_{hkl} for a unit cell containing atoms at fractional coordinates $\mathbf{r}_j = (x_j, y_j, z_j)$ is defined as:

$$F_{hkl} = \sum_{j=1}^{N} f_j \exp\left[-2\pi i(hx_j + ky_j + lz_j)\right]$$

where f_j is the atomic scattering form factor of the j-th atom.

[Marking Scheme: 1.0 Mark for definition and formula]

(ii) **FCC Lattice Derivation:** The conventional FCC unit cell contains 4 identical atoms ($f_j = f$) at fractional coordinates:

$$(0,0,0), \quad \left(\frac{1}{2},\frac{1}{2},0\right), \quad \left(\frac{1}{2},0,\frac{1}{2}\right), \quad \left(0,\frac{1}{2},\frac{1}{2}\right)$$

Substituting these positions into the structure factor expression:

$$F_{hkl} = f\left[1 + e^{-\pi i(h+k)} + e^{-\pi i(h+l)} + e^{-\pi i(k+l)}\right]$$

Since $e^{-i\pi n} = (-1)^n$:

- **Case 1: All h,k,l are even:** Then $h+k$, $h+l$, and $k+l$ are all even integers. $(-1)^{\text{even}} = +1$.

$$F_{hkl} = f[1+1+1+1] = 4f \implies |F_{hkl}|^2 = 16f^2$$

- **Case 2: All h,k,l are odd:** Sum of two odd numbers is even: $h+k, h+l, k+l$ are all even.

$$F_{hkl} = f[1+1+1+1] = 4f \implies |F_{hkl}|^2 = 16f^2$$

- **Case 3: Mixed indices (e.g., two even, one odd; or two odd, one even):** Two of the pairwise sums will be odd and one will be even (or vice versa).

$$F_{hkl} = f[1+1-1-1] = 0 \implies I_{hkl} \propto |F_{hkl}|^2 = 0$$

Thus, reflections are **observed** only if h,k,l are *unmixed* (all even or all odd). Mixed reflections vanish identically.

[Marking Scheme: 2.0 Marks for derivation + 1.0 Mark for extinction rule conclusion]

(b) Powder XRD of Gold (FCC, $a = 4.08$ Å, $\lambda = 1.5406$ Å) [3 Marks]

(i) **Lowest-Angle Reflections:** For FCC, allowed (hkl) in order of increasing $h^2+k^2+l^2$:

- First peak: (111) (all odd, $h^2 + k^2 + l^2 = 3$)
- Second peak: (200) (all even, $h^2 + k^2 + l^2 = 4$)
- Note: (100) and (110) are mixed and forbidden.

[Marking Scheme: 1.0 Mark for identifying (111) and (200)]

(ii) **Interplanar Spacings and Bragg Angles:** For a cubic crystal, $d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$, and Bragg's law is $\lambda = 2d_{hkl} \sin\theta$.

- **For** (111) **plane:**

$$d_{111} = \frac{4.08 \text{ Å}}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{4.08}{\sqrt{3}} \approx 2.3556 \text{ Å}$$

$$\sin\theta_{111} = \frac{\lambda}{2d_{111}} = \frac{1.5406}{2 \times 2.3556} \approx 0.3270 \implies \boldsymbol{\theta_{111} \approx 19.09^\circ} \quad (2\theta \approx 38.18^\circ)$$

- **For** (200) **plane:**

$$d_{200} = \frac{4.08 \text{ Å}}{\sqrt{2^2 + 0 + 0}} = \frac{4.08}{2} = 2.0400 \text{ Å}$$

$$\sin\theta_{200} = \frac{\lambda}{2d_{200}} = \frac{1.5406}{2 \times 2.0400} \approx 0.3776 \implies \boldsymbol{\theta_{200} \approx 22.18^\circ} \quad (2\theta \approx 44.37^\circ)$$

[Marking Scheme: 1.0 Mark for d_{hkl} + 1.0 Mark for θ values]

Question 2: Phonon Dynamics & Free Electron Gas [8 Marks]

(a) Thermal Conductivity and Phonon Scattering Mechanisms [5 Marks]

(i) **Temperature Dependence of $\kappa_{ph}(T) = \frac{1}{3}C_v v \ell_{ph}$:**

- **Low Temperatures ($T \to 0$ K, $T \ll \Theta_D$):** Phonon population is very small; phonon-phonon scattering is negligible. The phonon mean free path ℓ_{ph} is limited only by sample boundaries / crystal size (L), so $\ell_{ph} \approx$ constant. The sound velocity v is independent of T. Hence, $\kappa_{ph}(T)$ follows the specific heat: $\kappa_{ph} \propto C_v \propto T^3$.
- **Intermediate Temperatures ($T \sim 0.1\,\Theta_D$):** C_v continues to rise toward Dulong-Petit saturation, while Umklapp scattering begins to turn on with an activation behavior $\ell_{ph} \propto e^{\Theta_D/(bT)}$. The product $C_v \ell_{ph}$ goes through a prominent **thermal conductivity maximum**.
- **High Temperatures ($T \gg \Theta_D$):** Specific heat saturates to the classical Dulong-Petit constant value $3Nk_B$. The number of thermal phonons excited is proportional to T ($n_{ph} \propto k_B T/\hbar\omega$). The phonon collision rate is proportional to the number of scatterers, so $\ell_{ph} \propto 1/T$. Consequently, $\kappa_{ph} \propto 1/T$.

[Marking Scheme: 2.5 Marks for explaining the three regimes and identifying the peak]

(ii) **Normal (N) vs Umklapp (U) Processes:**

- In a 3-phonon collision: $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_3 + \mathbf{G}$.
- **Normal (N) Process ($\mathbf{G} = 0$):** $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_3$. Crystal momentum is conserved. Energy is redistributed among phonon modes, but the total heat current $\mathbf{J}_Q = \sum \hbar \mathbf{q} v_g$ is conserved. Therefore, N-processes alone cannot degrade heat current or produce thermal resistance.
- **Umklapp (U) Process ($\mathbf{G} \neq 0$):** The vector sum $\mathbf{q}_1 + \mathbf{q}_2$ extends outside the first Brillouin zone and is folded back by a non-zero reciprocal lattice vector $\mathbf{G}$. The resulting wavevector $\mathbf{q}_3 = \mathbf{q}_1 + \mathbf{q}_2 - \mathbf{G}$ has a component opposite to the original direction of energy flow. This destroys net phonon momentum, establishing a finite steady-state temperature gradient and non-zero thermal resistivity ($W_{ph} = 1/\kappa_{ph} > 0$).

[Marking Scheme: 2.5 Marks for vector equations, role of G, and momentum degradation]

(b) Wiedemann-Franz Law and Lorenz Number [3 Marks]

(i) **Wiedemann-Franz Law and Lorenz Number:** The ratio of electronic thermal conductivity κ_e to electrical conductivity σ is directly proportional to absolute temperature T:

$$\frac{\kappa_e}{\sigma T} = L \quad \text{(Lorenz Number)}$$

From Sommerfeld quantum theory: $\sigma = \frac{ne^2\tau}{m}$ and $\kappa_e = \frac{1}{3}C_{ve}v_F^2\tau = \frac{\pi^2}{3}\frac{nk_B^2 T\tau}{m}$. Dividing the two:

$$L = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2$$

Numerical evaluation:

$$L = \frac{\pi^2}{3}\left(\frac{1.38065 \times 10^{-23}\ \text{J/K}}{1.60218 \times 10^{-19}\ \text{C}}\right)^2 \approx \mathbf{2.443 \times 10^{-8}}\ \text{W} \cdot \mathbf{\Omega} \cdot \text{K}^{\mathbf{-2}} \quad (\text{or } \text{V}^2/\text{K}^2)$$

[Marking Scheme: 1.5 Marks for formulation and correct value of L]

(ii) **Thermal Conductivity of Copper at 300 K:** Given $\sigma = 5.90 \times 10^7\ \Omega^{-1} \cdot \text{m}^{-1}$ and $T = 300$ K:

$$\kappa_e = L\sigma T = (2.443 \times 10^{-8}\ \text{W} \cdot \Omega \cdot \text{K}^{-2}) \times (5.90 \times 10^7\ \Omega^{-1} \cdot \text{m}^{-1}) \times (300\ \text{K})$$

$$\kappa_{\mathbf{e}} \approx \mathbf{432.4}\ \text{W} \cdot \text{m}^{\mathbf{-1}} \cdot \text{K}^{\mathbf{-1}}$$

[Marking Scheme: 1.5 Marks for calculation and correct units]

PART B: POST-MIDTERM SOLUTIONS [35 MARKS]

Question 3: Periodic Potentials, Bloch Theorem & Band Theory [10 Marks]

(a) Nearly Free Electron (NFE) Model and Band Gap [6 Marks]

(i) **Bloch's Theorem:** The eigenstates $\psi_k(x)$ of the one-electron Schrödinger equation in a periodic potential $V(x) = V(x+a)$ are of the form:

$$\psi_k(x) = e^{ikx}u_k(x), \quad \text{where } u_k(x+a) = u_k(x) \text{ has lattice periodicity.}$$

Equivalently, $\psi_k(x+a) = e^{ika}\psi_k(x)$.

[Marking Scheme: 1.0 Mark]

(ii) **Two-Band Secular Equation Setup:** Substitute trial wavefunction $\psi(x) = C_1 e^{ikx} + C_2 e^{i(k-G)x}$ into $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$. Writing $V(x) = 2V_1\cos(Gx) = V_1 e^{iGx} + V_1 e^{-iGx}$ (with $V_1 = V_G = V_{-G}$ real), and taking inner products with $\langle e^{ikx}|$ and $\langle e^{i(k-G)x}|$:

$$\left(\frac{\hbar^2 k^2}{2m} - E\right) C_1 + V_1 C_2 = 0$$
$$V_1 C_1 + \left(\frac{\hbar^2 (k-G)^2}{2m} - E\right) C_2 = 0$$

For non-trivial coefficients $(C_1, C_2) \neq 0$, the determinant must vanish:

$$\begin{vmatrix} E_0(k) - E & V_1 \\ V_1 & E_0(k-G) - E \end{vmatrix} = 0$$

where $E_0(k) = \frac{\hbar^2 k^2}{2m}$ is the free-electron kinetic energy.

[Marking Scheme: 2.0 Marks for setting up matrix]

(iii) **Band Gap Opening at Zone Boundary:** At the zone boundary $k = G/2 = \pi/a$, we have $k - G = -G/2$, so the kinetic energies are equal:

$$E_0(G/2) = E_0(-G/2) = \frac{\hbar^2 (G/2)^2}{2m} = E_B$$

The secular equation simplifies to:

$$(E_B - E)^2 - V_1^2 = 0 \implies E_B - E = \pm V_1 \implies \mathbf{E_\pm = E_B \pm |V_1|}$$

The lower standing-wave state has energy $E_- = E_B - |V_1|$ (probability density piles up at the attractive ion cores), while the upper standing-wave state has energy $E_+ = E_B + |V_1|$ (probability density piles up between ions). The forbidden energy gap is:

$$\mathbf{E_g = E_+ - E_- = 2|V_1|}$$

[Marking Scheme: 2.0 Marks for solving eigenvalues and finding E_g]

(iv) **Reduced Zone Scheme Sketch:** The band structure consists of two energy branches $E_1(k)$ and $E_2(k)$ within $-\pi/a \leq k \leq +\pi/a$. At $k = 0$, $E_1(0) = 0$; at $k = \pm\pi/a$, $E_1(\pm\pi/a) = E_B - |V_1|$ and $E_2(\pm\pi/a) = E_B + |V_1|$, with zero group velocity ($dE/dk = 0$) at zone boundaries.

[Marking Scheme: 1.0 Mark for correct sketch]

(b) Tight-Binding Model (1D Chain) [4 Marks]

(i) **Dispersion Derivation:** Representing the Bloch state as $\psi_k(x) = \frac{1}{\sqrt{N}} \sum_n e^{ikna} \phi(x - na)$:

$$E(k) = \frac{\langle \psi_k | \hat{H} | \psi_k \rangle}{\langle \psi_k | \psi_k \rangle} = \epsilon_0 - te^{ika} - te^{-ika} = \boldsymbol{\epsilon_0 - 2t\cos(ka)}$$

where $\epsilon_0 = \langle \phi_n | \hat{H} | \phi_n \rangle$ and $t = -\langle \phi_n | \hat{H} | \phi_{n\pm1} \rangle > 0$.

[Marking Scheme: 2.0 Marks]

(ii) **Total Bandwidth W:**

- Minimum energy at $k = 0$: $E_{min} = \epsilon_0 - 2t$.
- Maximum energy at $k = \pi/a$: $E_{max} = \epsilon_0 + 2t$.
- Bandwidth: $\mathbf{W = E_{max} - E_{min} = 4t}$.

[Marking Scheme: 1.0 Mark]

(iii) **Effective Mass m^*:** Differentiating: $\frac{dE}{dk} = 2ta\sin(ka)$, and $\frac{d^2E}{dk^2} = 2ta^2\cos(ka)$.

$$m^*(k) = \frac{\hbar^2}{d^2E/dk^2} = \frac{\hbar^2}{2ta^2\cos(ka)}$$

- At band bottom ($k = 0$): $\cos(0) = +1 \implies \mathbf{m^*(0) = \frac{\hbar^2}{2ta^2}} > 0$ (electron-like).
- At band top ($k = \pi/a$): $\cos(\pi) = -1 \implies \mathbf{m^*(\pi/a) = -\frac{\hbar^2}{2ta^2}} < 0$ (negative effective mass).

[Marking Scheme: 1.0 Mark]

Question 4: Band Dynamics & Semiconductors [8 Marks]

(a) Band Dynamics, Effective Mass, and Holes [4 Marks]

(i) **Effective Mass Derivation:** Work done on an electron in wave packet by field $\mathcal{E}$ in time dt:

$$dE = Fv_g dt = (-e\mathcal{E})v_g dt = \hbar \frac{dk}{dt} v_g dt = \hbar v_g dk \implies \frac{dk}{dt} = \frac{F}{\hbar}$$

Differentiating group velocity $v_g = \frac{1}{\hbar}\frac{dE}{dk}$ with respect to time:

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d}{dt}\left(\frac{dE}{dk}\right) = \frac{1}{\hbar}\frac{d^2E}{dk^2}\frac{dk}{dt} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}F$$

Comparing with Newton's second law $F = m^*a$, we obtain:

$$\mathbf{m^* = \hbar^2 \left(\frac{d^2E}{dk^2}\right)^{-1}}$$

[Marking Scheme: 1.5 Marks]

(ii) **Negative Effective Mass and Concept of Holes:** Near the top of the band, the dispersion curve is convex downward ($\frac{d^2E}{dk^2} < 0$), giving $m^* < 0$. Physically, Bragg reflection from the lattice planes produces an internal reaction force from the periodic potential that exceeds the external field force, causing the electron to accelerate in the opposite direction. A full band carries zero net current. Removing an electron of wavevector $\mathbf{k}_e$, energy E_e, and charge $-e$ leaves a vacancy with:

$$\mathbf{k}_h = -\mathbf{k}_e, \quad E_h = -E_e, \quad q_h = +e, \quad m_h^* = -m_e^* > 0$$

This vacancy is treated as a quasiparticle called a **hole**.

[Marking Scheme: 1.5 Marks]

(iii) **Bloch Oscillations:** An electron in an electric field moves across k-space at rate $\hbar\dot{k} = -e\mathcal{E}$. As it reaches the Brillouin zone boundary, it undergoes Bragg reflection to $-G/2$, executing periodic oscillations in real and k-space with frequency $\omega_B = \frac{e\mathcal{E}a}{\hbar}$. In bulk metals, the electron scattering time $\tau \sim 10^{-14}$ s is far shorter than the oscillation period ($\tau \ll T_B = \frac{2\pi\hbar}{e\mathcal{E}a}$), so electrons scatter long before completing an oscillation.

[Marking Scheme: 1.0 Mark]

(b) Semiconductor Carrier Statistics & Silicon Calculation [4 Marks]

(i) **Intrinsic Carrier Concentration $n_i(T)$:** For non-degenerate carrier statistics ($E_c - E_F \gg k_BT$ and $E_F - E_v \gg k_BT$):

$$n = N_c e^{-(E_c - E_F)/k_BT}, \quad p = N_v e^{-(E_F - E_v)/k_BT}$$

In an intrinsic material, $n = p = n_i$. Multiplying the two (Law of Mass Action):

$$n_i^2 = np = N_c N_v \exp\left(-\frac{E_c - E_v}{k_BT}\right) = N_c N_v \exp\left(-\frac{E_g}{k_BT}\right)$$

Taking the square root gives: $\mathbf{n_i(T) = \sqrt{N_c N_v}\, e^{-E_g/(2k_BT)}}$.

[Marking Scheme: 1.5 Marks]

(ii) **Intrinsic Fermi Level E_{Fi}:** Equating $n = p$:

$$N_c e^{-(E_c - E_{Fi})/k_B T} = N_v e^{-(E_{Fi} - E_v)/k_B T} \implies e^{2E_{Fi}/k_B T} = \frac{N_v}{N_c} e^{(E_c + E_v)/k_B T}$$

Taking the natural logarithm:

$$E_{Fi} = \frac{E_c + E_v}{2} + \frac{1}{2} k_B T \ln\left(\frac{N_v}{N_c}\right) = \frac{\mathbf{E_c + E_v}}{\mathbf{2}} + \frac{\mathbf{3}}{\mathbf{4}} \mathbf{k_B T} \ln\left(\frac{\mathbf{m_h^*}}{\mathbf{m_e^*}}\right)$$

[Marking Scheme: 1.0 Mark]

(iii) **Numerical Calculation for Silicon ($T = 300$ K):** $k_B T = 1.38065 \times 10^{-23} \times 300 / 1.60218 \times 10^{-19} \approx 0.02585$ eV.

$$N_c = 2\left(\frac{2\pi \times 1.08 \times 9.109 \times 10^{-31} \times 1.38065 \times 10^{-23} \times 300}{(6.626 \times 10^{-34})^2}\right)^{3/2} \approx \mathbf{2.82 \times 10^{25}}\ \mathrm{m^{-3}} \quad (2.82{\times}10^{19}\ \mathrm{cm^{-3}})$$

$$N_v = N_c \left(\frac{m_h^*}{m_e^*}\right)^{3/2} = 2.82 \times 10^{25} \times \left(\frac{0.56}{1.08}\right)^{3/2} \approx \mathbf{1.05 \times 10^{25}}\ \mathrm{m^{-3}} \quad (1.05 \times 10^{19}\ \mathrm{cm^{-3}})$$

$$n_i = \sqrt{N_c N_v} e^{-E_g/2k_B T} = \sqrt{2.82 \times 1.05 \times 10^{50}} \exp\left(-\frac{1.12}{2 \times 0.02585}\right)$$

$$n_i \approx 1.72 \times 10^{25} \times e^{-21.66} \approx \mathbf{6.73 \times 10^{15}}\ \mathrm{m^{-3}} \quad (\mathbf{6.73 \times 10^{9}}\ \mathrm{cm^{-3}})$$

Fermi level shift from midgap:

$$\Delta E_{Fi} = \frac{3}{4} k_B T \ln\left(\frac{0.56}{1.08}\right) = 0.75 \times 25.85\ \mathrm{meV} \times (-0.6568) \approx \mathbf{-12.73}\ \mathrm{meV}$$

The intrinsic Fermi level lies 12.7 meV below midgap because holes are lighter than electrons ($m_h^* < m_e^*$).

[Marking Scheme: 1.5 Marks]

Question 5: Dielectrics & Optical Properties [9 Marks]

(a) Local Field and Clausius-Mossotti Derivation [5 Marks]

(i) **Macroscopic vs Local Electric Field:** The macroscopic field $\mathbf{E}$ is the spatial average of the microscopic electric field over many unit cells. The local field $\mathbf{E}_{loc}$ is the actual field experienced by a single atom at its specific site, excluding the field produced by the atom itself.

[Marking Scheme: 1.0 Mark]

(ii) **Lorentz Local Field Derivation:** Construct a spherical Lorentz cavity around the reference site. The local field is composed of:

$$\mathbf{E}_{loc} = \mathbf{E} + \mathbf{E}_1 + \mathbf{E}_2$$

where $\mathbf{E}$ is the applied macroscopic field (including plate polarization), $\mathbf{E}_1$ is the field due to surface polarization charge on the spherical cavity, and $\mathbf{E}_2$ is the sum of dipole fields inside the sphere. For cubic symmetry, $\mathbf{E}_2 = 0$. Surface charge density on cavity surface is $\sigma_{pol} = -P\cos\theta$. Integrating over the sphere:

$$\mathbf{E}_1 = \int \frac{\sigma_{pol}\cos\theta}{4\pi\epsilon_0 R^2} dA = \frac{P}{4\pi\epsilon_0}\int_0^\pi \cos^2\theta(2\pi\sin\theta d\theta) = \frac{\mathbf{P}}{3\epsilon_0}$$

Therefore, $\mathbf{E}_{loc} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}$.

[Marking Scheme: 1.5 Marks]

(iii) **Clausius-Mossotti Relation:** Polarization is related to atomic polarizability α by $\mathbf{P} = N\alpha\mathbf{E}_{loc} = N\alpha\left(\mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}\right)$. Solving for $\mathbf{P}$:

$$\mathbf{P}\left(1 - \frac{N\alpha}{3\epsilon_0}\right) = N\alpha\mathbf{E} \implies \mathbf{P} = \frac{N\alpha}{1 - \frac{N\alpha}{3\epsilon_0}}\mathbf{E}$$

Using $\mathbf{D} = \epsilon_0\epsilon_r\mathbf{E} = \epsilon_0\mathbf{E} + \mathbf{P} \implies \mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E}$:

$$\epsilon_0(\epsilon_r - 1) = \frac{N\alpha}{1 - \frac{N\alpha}{3\epsilon_0}} \implies \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$$

[Marking Scheme: 1.5 Marks]

(iv) **Polarization Catastrophe:** When the condition $\frac{N\alpha}{3\epsilon_0} \to 1$ is approached, the denominator diverges, meaning $\epsilon_r \to \infty$. This instability causes a spontaneous polarization $\mathbf{P} \neq 0$ even in zero external field, driving the material into the **ferroelectric phase**.

[Marking Scheme: 1.0 Mark]

(b) Plasma Oscillations and Optical Properties [4 Marks]

(i) **Dielectric Function Derivation:** Equation of motion for an undamped electron: $m_e\frac{d^2x}{dt^2} = -eE_0e^{-i\omega t}$. Assuming $x(t) = x_0e^{-i\omega t}$: $-m_e\omega^2 x = -eE \implies x = \frac{eE}{m_e\omega^2}$. The dipole moment per unit volume is $P = -nex = -\frac{ne^2E}{m_e\omega^2}$. Displacement: $D = \epsilon_0E + P = \epsilon_0\left(1 - \frac{ne^2}{\epsilon_0 m_e\omega^2}\right)E = \epsilon_0\epsilon(\omega)E$. Hence:

$$\epsilon(\omega) = \mathbf{1} - \frac{\omega_\mathbf{p}^\mathbf{2}}{\omega^\mathbf{2}}, \quad \text{with } \omega_\mathbf{p} = \sqrt{\frac{\mathbf{ne^2}}{\epsilon_\mathbf{0}\mathbf{m_e}}}$$

[Marking Scheme: 1.5 Marks]

(ii) **Reflectance and Transparency:** Refractive index $\tilde{n} = \sqrt{\epsilon(\omega)}$.

- For $\omega < \omega_p$: $\epsilon(\omega) < 0 \implies \tilde{n} = i\kappa$ is purely imaginary. Light waves decay exponentially inside the metal (evanescent skin effect). Normal reflectance $R = \left|\frac{\tilde{n}-1}{\tilde{n}+1}\right|^2 = \frac{\kappa^2+1}{\kappa^2+1} = 1.0$. The metal is **totally reflecting / shiny**.
- For $\omega > \omega_p$: $\epsilon(\omega) > 0 \implies \tilde{n}$ is real. Light propagates through the medium without attenuation. The metal becomes **transparent** (ultraviolet transparency).

[Marking Scheme: 1.5 Marks]

(iii) **Numerical Calculation ($n = 6.00 \times 10^{28}$ $\mathbf{m^{-3}}$):**

$$\omega_p = \sqrt{\frac{(6.00 \times 10^{28}) \times (1.60218 \times 10^{-19})^2}{8.85419 \times 10^{-12} \times 9.10938 \times 10^{-31}}} = \sqrt{\frac{1.5402 \times 10^{-9}}{8.0656 \times 10^{-42}}} \approx \mathbf{1.382 \times 10^{16}} \text{ rad} \cdot \text{s}^{-1}$$

$$\nu_p = \frac{\omega_p}{2\pi} = \frac{1.382 \times 10^{16}}{2\pi} \approx \mathbf{2.20 \times 10^{15}} \text{ Hz}$$

$$\lambda_p = \frac{c}{\nu_p} = \frac{2.998 \times 10^8 \text{ m/s}}{2.20 \times 10^{15} \text{ s}^{-1}} \approx 1.363 \times 10^{-7} \text{ m} = \mathbf{136.3} \text{ nm} \quad \text{(Vacuum Ultraviolet)}$$

[Marking Scheme: 1.0 Mark]

Question 6: Magnetism & Exchange Interaction [8 Marks]

(a) Weiss Molecular Field Theory & Curie-Weiss Law [4 Marks]

(i) **Molecular Field Hypothesis:** Weiss assumed that atomic magnetic moments interact through an effective internal magnetic field B_{eff} proportional to the magnetization:

$$B_{eff} = B_a + \lambda M$$

where B_a is the external field and λ is the Weiss molecular field parameter.

[Marking Scheme: 1.0 Mark]

(ii) **Derivation of Curie-Weiss Law:** Above the transition temperature, moments follow Curie's law with respect to B_{eff}:

$$M = \frac{C}{T} B_{eff} = \frac{C}{T}(B_a + \lambda M)$$

Rearranging terms:

$$M\left(1 - \frac{\lambda C}{T}\right) = \frac{C}{T} B_a \implies M\left(\frac{T - \lambda C}{T}\right) = \frac{C}{T} B_a \implies M = \frac{C B_a}{T - \lambda C}$$

The magnetic susceptibility $\chi = M/B_a$ is therefore:

$$\chi = \frac{\mathbf{C}}{\mathbf{T} - \theta_{\mathbf{C}}}, \quad \text{with } \theta_{\mathbf{C}} = \lambda \mathbf{C}$$

This is the **Curie-Weiss law**. The singularity at $T = \theta_C = T_C$ marks the onset of ferromagnetic order.

[Marking Scheme: 1.5 Marks]

(iii) **Spontaneous Magnetization below T_C:** In zero applied field ($B_a = 0$), $B_{eff} = \lambda M$. The magnetization satisfies the self-consistent relation:

$$\frac{M}{M_s} = \tanh\left(\frac{g\mu_B S \lambda M}{k_B T}\right) \quad \text{or } m = \tanh\left(\frac{T_C}{T} m\right)$$

Plotting $y_1 = m$ and $y_2 = \tanh((T_C/T)m)$ against m:

- For $T > T_C$, initial slope $\frac{T_C}{T} < 1$; the only intersection is at $m = 0$ (paramagnetic state).
- For $T < T_C$, initial slope $\frac{T_C}{T} > 1$; a non-zero stable root $m_0 \neq 0$ exists, corresponding to **spontaneous magnetization**.

[Marking Scheme: 1.5 Marks]

(b) Quantum Exchange Interaction & Antiferromagnetism [4 Marks]

(i) **Failure of Dipolar Interactions:** The interaction energy between two Bohr magnetons separated by interatomic distance $a \approx 3$ Å is:

$$E_{dip} \sim \frac{\mu_0 \mu_B^2}{4\pi a^3} \approx 10^{-23} \text{ J} \implies T_{dip} \sim \frac{E_{dip}}{k_B} \approx 1 \text{ K}$$

Classical dipolar coupling predicts ferromagnetic ordering only below ~ 1 K, failing by a factor of 10^3 to explain real materials like Iron ($T_C = 1043$ K).

[Marking Scheme: 1.0 Mark]

(ii) **Heisenberg Exchange Interaction:** For a two-electron system, the spatial wavefunction must be symmetric (antisymmetric) when the spin state is a singlet $S = 0$ (triplet $S = 1$). The difference in electrostatic Coulomb energy between singlet and triplet states is $E_{singlet} - E_{triplet} = 2J$, where J is the exchange integral:

$$J = \int \phi_A^*(\mathbf{r}_1)\phi_B^*(\mathbf{r}_2)\frac{e^2}{4\pi\epsilon_0|\mathbf{r}_1 - \mathbf{r}_2|}\phi_A(\mathbf{r}_2)\phi_B(\mathbf{r}_1)d^3r_1 d^3r_2$$

Using $\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2}[S_{tot}(S_{tot}+1) - S_1(S_1+1) - S_2(S_2+1)]$, this energy difference can be represented as the spin Hamiltonian:

$$\hat{H}_{ex} = -2J\mathbf{S}_1 \cdot \mathbf{S}_2 \quad \text{or for a lattice } \hat{H}_{ex} = -2\sum_{i<j} J_{ij}\mathbf{S}_i \cdot \mathbf{S}_j$$

- **Ferromagnetic ($J > 0$):** Parallel spin alignment ($\uparrow\uparrow$) minimizes energy.
- **Antiferromagnetic ($J < 0$):** Antiparallel spin alignment ($\uparrow\downarrow$) minimizes energy.

[Marking Scheme: 1.5 Marks]

(iii) **Antiferromagnetic Susceptibility $\chi(T)$:**

- For $T > T_N$: Susceptibility follows the Curie-Weiss law: $\chi = \frac{C}{T+\theta_N}$ (with positive θ_N).
- At $T = T_N$: Susceptibility reaches a maximum cusp.
- Below T_N:
  - **Perpendicular field ($\chi_\perp$):** Spins cant against exchange field; $\chi_\perp \approx$ constant independent of T.
  - **Parallel field ($\chi_\parallel$):** Spins cannot cant; thermal excitation is required to flip spins against the internal exchange field. As $T \to 0$ K, $\chi_\parallel \to 0$.

  [Marking Scheme: 1.5 Marks]

***** END OF COMPREHENSIVE SOLUTIONS MANUAL *****